Lecture 04-05

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1 Mathematical Induction

1.1 Introduction

Mathematical induction is a proof technique used to prove a statement $P(n)$ for all natural numbers $n \in \mathbb{N}$.

1.1.1 Analogy

Imagine a row of ducks (or birds).

1. The first duck flies away.
2. If any duck flies away, the next one also flies away.

Conclusion: All ducks fly away.

1.1.2 The Principle of Mathematical Induction

To prove that a statement $P(n)$ is true for all $n \geq 1$:

1. **Basis Step**: Prove that $P(1)$ is true.
2. **Inductive Step**: Assume that $P(n)$ is true (this is called the Inductive Hypothesis). Show that $P(n) \implies P(n+1)$.
   - Meaning: Assuming the statement holds for n, we must prove it holds for $n+1$.

1.2 Problem 1: Sum of Squares

Proposition: For all $n \in \mathbb{N}$:

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

1.2.1 Proof

1. **Basis**: Let $n = 1$.

$$1^2 = 1$$
$$\frac{1(1+1)(2(1)+1)}{6} = \frac{1 \cdot 2 \cdot 3}{6} = 1$$

   The statement is true for $n = 1$.

2. **Assumption**: Assume the statement is true for n.

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

3. **Inductive Step**: We must show it is true for $n+1$, i.e.,

$$\sum_{i=1}^{n+1} i^2 = \frac{(n+1)(n+2)(2n+3)}{6}$$

Derivation:

$$\text{LHS} = \left(\sum_{i=1}^{n} i^2\right) + (n+1)^2$$

Using the assumption:

$$= \frac{n(n+1)(2n+1)}{6} + (n+1)^2$$

Factor out $(n+1)$:

$$= (n+1)\left[\frac{n(2n+1)}{6} + \frac{6(n+1)}{6}\right]$$

$$= \frac{n+1}{6}[2n^2 + n + 6n + 6]$$

$$= \frac{n+1}{6}[2n^2 + 7n + 6]$$

Factoring the quadratic $2n^2 + 7n + 6 = (n+2)(2n+3)$:

$$= \frac{(n+1)(n+2)(2n+3)}{6} = \text{RHS}$$

Q.E.D.

1.3 Problem 2: Divisibility

Proposition: Prove that $6 \mid (n^3 + 5n)$ for all $n \in \mathbb{N}$. **(Notation: $a \mid b$ means "a divides b")**

1.3.1 Proof

1. **Basis**: Let $n = 1$.

$$1^3 + 5(1) = 6$$

Since $6 \mid 6$, the statement is true.

2. **Assumption**: Assume $6 \mid (n^3 + 5n)$. This implies $n^3 + 5n = 6k$ for some integer k.

3. **Inductive Step**: Consider $P(n+1)$. We examine $(n+1)^3+5(n+1)$.

$$(n+1)^3 + 5(n+1) = (n^3 + 3n^2 + 3n + 1) + (5n + 5)$$

Group the terms from the assumption $(n^3 + 5n)$:

$$= (n^3 + 5n) + 3n^2 + 3n + 6$$

$$= (n^3 + 5n) + 3n(n+1) + 6$$

Analysis of terms:

- $(n^3 + 5n)$ is divisible by 6 (by assumption).
- 6 is divisible by 6.
- Consider $3n(n+1)$: The term $n(n+1)$ is the product of two consecutive integers, so it is always even (divisible by 2).

$$n(n+1) = 2m \implies 3n(n+1) = 3(2m) = 6m$$

Thus, $3n(n+1)$ is divisible by 6.

Since all terms are divisible by 6, the sum is divisible by 6.

1.4 Problem 3: Sum of Odd Numbers

Conjecture: $1 + 3 + 5 + \cdots + (2n-1) = n^2$.

1.4.1 Proof

1. **Basis**: Let $n = 1$.

$$2(1) - 1 = 1$$

$$1^2 = 1$$

True.

2. **Assumption**: Assume $\sum_{i=1}^{n}(2i-1) = n^2$.

3. **Inductive Step**: Prove for $n+1$.

$$\sum_{i=1}^{n+1}(2i-1) = \sum_{i=1}^{n}(2i-1) + (2(n+1)-1)$$

Using the assumption:

$$= n^2 + (2n + 2 - 1)$$
$$= n^2 + 2n + 1$$
$$= (n + 1)^2$$

Q.E.D.

1.5 Problem 4: Inequality and Hypothesis Strengthening

Proposition: Prove $\sum_{i=1}^{n} \frac{1}{i^2} < 2$ for all $n \in \mathbb{N}$.

1.5.1 Attempt 1 (Standard Induction) - Fails

- **Basis**: $1 < 2$. True.
- **Assumption**: $\sum_{i=1}^{n} \frac{1}{i^2} < 2$.
- **Step**: $\sum_{i=1}^{n+1} \frac{1}{i^2} = \sum_{i=1}^{n} \frac{1}{i^2} + \frac{1}{(n+1)^2} < 2 + \frac{1}{(n+1)^2}$.
- **Issue**: This does not prove the sum is strictly less than 2. It shows it is less than something slightly larger than 2. The inductive hypothesis is too weak.

1.5.2 Attempt 2 (Strengthened Hypothesis)

New Conjecture: $\sum_{i=1}^{n} \frac{1}{i^2} \leq 2 - \frac{1}{n}$.
Proof:

1. **Basis**: $n = 1$.
$$\frac{1}{1^2} \leq 2 - \frac{1}{1} \implies 1 \leq 1$$
True.
2. **Assumption**: $\sum_{i=1}^{n} \frac{1}{i^2} \leq 2 - \frac{1}{n}$.
3. **Inductive Step**: Show $\sum_{i=1}^{n+1} \frac{1}{i^2} \leq 2 - \frac{1}{n+1}$. Using the assumption:
$$\sum_{i=1}^{n+1} \frac{1}{i^2} = \sum_{i=1}^{n} \frac{1}{i^2} + \frac{1}{(n+1)^2} \leq \left(2 - \frac{1}{n}\right) + \frac{1}{(n+1)^2}$$

We need to prove:

$$2 - \frac{1}{n} + \frac{1}{(n+1)^2} \leq 2 - \frac{1}{n+1}$$

Subtract 2 from both sides and rearrange:

$$\frac{1}{(n+1)^2} \leq \frac{1}{n} - \frac{1}{n+1}$$

$$\frac{1}{(n+1)^2} \leq \frac{n+1-n}{n(n+1)}$$

$$\frac{1}{(n+1)^2} \leq \frac{1}{n(n+1)}$$

Multiplying both sides by positive terms $(n+1)^2$ and $n(n+1)$ or simply observing denominators:

$$(n+1)^2 \geq n(n+1)$$

$$n^2 + 2n + 1 \geq n^2 + n$$

$$n + 1 \geq 0$$

This is clearly true for $n \in \mathbb{N}$. Thus, the inequality holds.

1.6 Problem 5: Triangle Inequality Generalization

Proposition: For all n, $|X_1 + \cdots + X_n| \leq |X_1| + \cdots + |X_n|$.

1.6.1 Proof

1. **Basis**: $n = 1$ is trivial ($|X_1| \leq |X_1|$). $n = 2$ is the standard Triangle Inequality: $|X_1 + X_2| \leq |X_1| + |X_2|$.

2. **Assumption**: Assume it holds for n.

$$|\sum_{i=1}^{n} X_i| \leq \sum_{i=1}^{n} |X_i|$$

3. **Inductive Step**:

$$|\sum_{i=1}^{n+1} X_i| = |(\sum_{i=1}^{n} X_i) + X_{n+1}|$$

Treat $(\sum_{i=1}^{n} X_i)$ as a single term A. By the standard triangle inequality ($n = 2$ case):

$$|A + X_{n+1}| \leq |A| + |X_{n+1}|$$
$$\leq |\sum_{i=1}^{n} X_i| + |X_{n+1}|$$

By the inductive assumption for the first term:

$$\leq (\sum_{i=1}^{n} |X_i|) + |X_{n+1}| = \sum_{i=1}^{n+1} |X_i|$$

Q.E.D.

1.7 Problem 6: Factorial Identity

Proposition: $\sum_{k=1}^{n} k \cdot k! = (n+1)! - 1$. **(Note:** $n! = 1 \cdot 2 \cdot \cdots \cdot n$ **and** $0! = 1$**)**

1.7.1 Proof Idea

1. **Basis**: $1 \cdot 1! = 1$. $(1+1)! - 1 = 2 - 1 = 1$. True.
2. **Step**:

$$\text{LHS for } n+1 = [(n+1)! - 1] + (n+1)(n+1)!$$
$$= (n+1)![1 + (n+1)] - 1$$
$$= (n+1)!(n+2) - 1$$
$$= (n+2)! - 1$$

Q.E.D.

1.8 Problem 7: Tiling with Triminoes

Problem: A $2^n \times 2^n$ chessboard with exactly one square removed can be tiled using L-shaped triminoes (tiles covering 3 squares).

1.8.1 Proof

1. **Basis** ($n = 1$): A 2×2 board has 4 squares. If we remove 1, we have 3 squares in an L-shape. This can be perfectly covered by one trimino.

2. **Assumption**: Assume any $2^n \times 2^n$ board with one square removed can be tiled.

3. **Inductive Step**: Consider a $2^{n+1} \times 2^{n+1}$ board with one missing square.

   - Divide the board into four $2^n \times 2^n$ quadrants.
   - The missing square is located in one of these quadrants.
   - By hypothesis, the quadrant with the missing square can be tiled.
   - For the other three quadrants, place a single trimino at the center of the large board such that it covers one corner square of each of the three "full" quadrants.
   - Now, each of these three quadrants has exactly one square "removed" (occupied by the center trimino).
   - By the inductive hypothesis, these three quadrants can also be tiled.
   - Thus, the entire board is tiled.

1.9 Additional Examples

1.9.1 Coin/Stamp Problem

Problem: One can form any amount $n \geq 8$ using only 3 and 5 euro coins/stamps. **Basis**:

- $8 = 3 + 5$
- $9 = 3 + 3 + 3$
- $10 = 5 + 5$

Step: If we can form k, we can form $k + 3$ by adding a 3 euro coin (Generalizing via strong induction or cases).